



Capital University of Science and Technology

Department of Software Engineering

SE3522 – Computer Org & Assembly Language

Assignment # 3

Semester: Fall 2025

Max Marks:10

Date: Dec 28, 2025

Last date: Jan 03, 2025

Instructor: Imran ul haq

Instructions:

- Submit a hard copy of your work within the specified time.
(Jan 03, 8:00 PM Eve)
 - Attempt all questions neatly and show complete workings where required.
 - Each part carries equal marks.
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Questions

Q1. Addressing Modes & Size Mismatch (5 Marks)

Consider the following data declarations and code snippet written for an 8086 processor:

DATA SEGMENT

num1 DB 45h

num2 DW 1234h

array DB 10h, 20h, 30h, 40h

DATA ENDS

CODE SEGMENT

MOV AX, DATA

MOV DS, AX

```
MOV AL, num1
MOV BX, OFFSET num2
MOV AX, [BX]
MOV CL, [array + 2]
MOV AX, num1
CODE ENDS
```

- Identify **two instructions** that will cause a **size mismatch or logical error**.
- Explain **why** each error occurs.
- Rewrite the incorrect instructions using **valid addressing and correct operand sizes**.

Q2. Register Indirect, Offset Addressing & Wraparound (5 Marks)

Assume the following situation in an 8086 system:

- DS = 2000h
- SI = FFFEh
- A byte array starts at offset FFFAh

```
MOV AL, [SI]
INC SI
MOV BL, [SI]
```

- Determine the **physical address accessed** by each instruction.
- Explain how **address wraparound** occurs in this case.
- State whether the wraparound is handled by the **programmer or the processor**, and justify your answer.

--- Good Luck ---